

# Icellmobilsoft wildfly dockerimage

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## 1. Common

The repository is a collection of wildfly docker images used in icellmobilsoft, to keep projects and their builds on the same basis.

## 2. Dockerfile

The dockerfile structure is based on onion layering, can be reused to build another docker image. To do this, you need to give it the base image you want and it will add to it as needed. The required base images must meet certain conditions, which are usually bound by the names of the ENV parameters. These conditions are already described in [Dockerhub Icellmobilsoft image](#) repository.

### *LABEL*

LABEL is a very important part of the builded docker images. They show all the data, that was an important factor in building the docker image. A sample of how this can be extracted from the docker image:

```
<CONSOLE>:~$ docker inspect icellmobilsoft/wildfly-full/oracle-jdbc-ibm-mq:1.0.0-SNAPSHOT | jq -r '.[0].Config.Labels' ❶
{
  "apt-get.install": "fontconfig,fonts-dejavu", ❷
  "base.image": "icellmobilsoft/wildfly-full/oracle-jdbc:1.0.0-SNAPSHOT", ❸
  "builder.images": "icellmobilsoft.hu/icellmobilsoft/builder-java17-galleon:1.1.0-SNAPSHOT", ❹
  "ee.configs": "managed-executor",
  "galleon.image": "icellmobilsoft/icellmobilsoft/builder-java17-galleon:1.1.0-SNAPSHOT", ❺
  "galleon.layers": "full", ❻
  "ibm-mq.version": "com.ibm.mq:wmq.jmsra:9.1.0.8",
  "image.usage": "release", ❼
  "java.image": "eclipse-temurin:17.0.7-jre", ❽
  "logstash.version": "biz.paluch.logging:logstash-gelf:1.14.1",
  "microprofile.configs": "opentrace,metrics",
  "oracle-jdbc.version":
"com.oracle.database.jdbc:ojdbc10:19.7.0.0,com.oracle.database.nls:orai18n:19.7.0.0",
  "wildfly.modules": "logstash,oracle-jdbc,ibm-mq", ❾
  "wildfly.properties.file": "/home/icellmobilsoft/config/wildfly.properties", ❿
  "wildfly.version": "27.0.1.Final" ⓫
}
```

- ❶ the command itself that cuts the label part from the complete docker inspect response
- ❷ programs installed on linux base image
- ❸ base image from which this particular image is derived (onion layering)
- ❹ the images used in the build
- ❺ name of the image that did the wildfly build
- ❻ which galleon modules are used to build wildfly
- ❼ image purpose. Intended for controlling, application image can only contain "release" value.
- ❽ what image the java in the image comes from
- ❾ what modules are manually installed in wildfly
- ❿ the properties file used to start wildfly
- ⓫ wildfly version

Some of the labels are from base image, when creating them, you have to be careful when to overwrite them.

## 2.1. wildfly/common

In terms of layers, this is the only layer where the galleon is used to put together the wildfly application server. The base images are complemented with the logger java module serving logstash. For more information on layers, see the documentation at [wildfly\\_galleon\\_layers](#).

In this case we will create these images from it:

- **icellmobilsoft/wildfly-default-ms:\${VERSION}** - Image meets basic microservice requirements
- **icellmobilsoft/wildfly-default-jpa:\${VERSION}** - Add jpa layer to the microservice image above
- **icellmobilsoft/wildfly-default-full:\${VERSION}** - Full wildfly with all existing modules

The dockerfile is responsible for EXPOSE of the basic 8080 and 9990 (+ others) ports, and the wildfly startup CMD.

Table 1. Configuration options:

| ENV kulcs                                 | wildfly.properties kulcs                  | Default | Leírás                                                                                                  |
|-------------------------------------------|-------------------------------------------|---------|---------------------------------------------------------------------------------------------------------|
| LOGGING_ROOT_LOGGER_LOGLEVEL              | logging.root.logger.loglevel              | INFO    | Global log level                                                                                        |
| LOGGING_HU_ICELLMOBILSOFT_LOGGER_LOGLEVEL | logging.hu.icellmobilsoft.logger.loglevel | ALL     | Global log level for "hu.icellmobilsoft" packages                                                       |
| LOGGING_COFFEE_ETCD_LOGGER_LOGLEVEL       | logging.coffee.etcd.logger.loglevel       | INFO    | Log level regarding ETCD classes                                                                        |
| LOGGING_K8S_NAMESPACE                     | -                                         | -       | Value of key <b>k8s_namespace</b> in the logged string (log pattern)                                    |
| CONSOLE_LOGGING_ENABLED                   | console.logging.enabled                   | true    | Enable plaintext console logging                                                                        |
| CONSOLE_LOGGING_JSON_ENABLED              | console.logging.json.enabled              | false   | Enable json console logging.                                                                            |
| JSON_MODULE_ID                            | json.module.id                            | unknown | Set the json logging additionalFields <b>moduleId</b> key value. Preferably pre-set in Dockerfile.      |
| JSON_MODULE_VERSION                       | json.module.version                       | unknown | Set the json logging additionalFields <b>moduleVersion</b> key value. Preferably pre-set in Dockerfile. |
| JSON_K8S_NAMESPACE                        | json.k8s.namespace                        | unknown | Set the json logging additionalFields <b>K8S_NAMESPACE</b> key value.                                   |
| CONSOLE_LOGGING_GELF_JSON_ENABLED         | console.logging.gelf.json.enabled         | false   | Enable gelf json console logging.                                                                       |

| ENV kulcs                | wildfly.properties kulcs | Default   | Leírás                                                                                                       |
|--------------------------|--------------------------|-----------|--------------------------------------------------------------------------------------------------------------|
| GELF_JSON_MODULE_ID      | gelf.json.module.id      | unknown   | Set the gelf json logging additionalFields <b>moduleId</b> key value. Preferably pre-set in Dockerfile.      |
| GELF_JSON_MODULE_VERSION | gelf.json.module.version | unknown   | Set the gelf json logging additionalFields <b>moduleVersion</b> key value. Preferably pre-set in Dockerfile. |
| GELF_JSON_K8S_NAMESPACE  | gelf.json.k8s.namespace  | unknown   | Set the gelf json logging additionalFields <b>K8S_NAMESPACE</b> key value.                                   |
| GELF_JSON_MDC_FIELDS     | gelf.json.mdc.fields     | -         | Set the gelf json logging additionalFields additional key values.                                            |
| FILE_LOGGING_ENABLED     | file.logging.enabled     | false     | Enable file logging. The maximum file size is 50MB by default.                                               |
| LOGSTASH_ENABLED         | logstash.enabled         | false     | Enable sending log entries to logstash server                                                                |
| LOGSTASH_BLOCKING_MODE   | logstash.blocking.mode   | DISCARD   | Enable (BLOCK) or disable (DISCARD) blocking mode in case logs cannot be written                             |
| LOGSTASH_QUEUE_LENGTH    | logstash.queue.length    | 50000     | Size of the async queue length                                                                               |
| LOGSTASH_HOST            | logstash.host            | 127.0.0.1 | Logstash server host.                                                                                        |
| LOGSTASH_PORT            | logstash.port            | 8000      | Logstash server port.                                                                                        |
| LOGSTASH_MODULE_ID       | -                        | unknown   | Set the logstash logging additionalFields <b>moduleId</b> key value.                                         |
| LOGSTASH_MODULE_VERSION  | -                        | unknown   | Set the logstash logging additionalFields <b>moduleVersion</b> key value.                                    |

| ENV kulcs               | wildfly.properties kulcs | Default      | Leírás                                                                                                                                                                                                            |
|-------------------------|--------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LOGSTASH_K8S_NAME SPACE | -                        | unknown      | Set the logstash logging additionalFields <b>K8S_NAMESPACE</b> key value.                                                                                                                                         |
| LOGSTASH_MDC_FIELDS     | -                        | -            | Set the logstash logging additionalFields additional key values.                                                                                                                                                  |
| JAVA_GC                 | -                        | -XX:+UseG1GC | Set g1 garbage collector for default gc. You can change the gc via filling with different value: '-XX:+UseConcMarkSweepGC' Or set to empty string for JVM default gc. Example: <a href="#">docker-compose.yml</a> |

#### NOTE

The value of the key defined in **wildfly.properties** has been overridden by the value of the key defined in **ENV**

## 2.2. wildfly/microprofile

Expands the base image created by [wildfly/common](#) with microprofile, executor and other config. This is necessary because wildfly builds the standalone.xml differently depending on the galleon layer. This layer combines all mp and ee configs that are good for full and ms wildfly.

In this case we create these images from it:

- **icellmobilsoft/wildfly-mp-ms:\${VERSION}** - Basic microservice + microprofile configs
- **icellmobilsoft/wildfly-mp-jpa:\${VERSION}** - Add jpa layer to the microservice image above
- **icellmobilsoft/wildfly-mp-full:\${VERSION}** - Full wildfly with all existing modules + microprofile configs

All images above contain a separate managed-executor-service definition (java:jboss/ee/concurrency/executor/grpc) for gRPC calls and the core threads are configurable as shown below.

Configuration options are inherited from the parent [wildfly/common](#) image and supplemented with the following:

*Table 2. Configuration options:*

| ENV kulcs                                  | wildfly.properties kulcs                  | Default                                                     | Leírás                                                                                                                                                                                                                                                             |
|--------------------------------------------|-------------------------------------------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MANAGED_EXECUTOR_SERVICE_CORE_THREADS      | managed-executor-service.core-thread      | 16                                                          | Minimum number of Wildfly managed executors. (If it was not configured, it would be dynamic - [number of cpu cores] x 2)                                                                                                                                           |
| MANAGED_EXECUTOR_SERVICE_GRPC_CORE_THREADS | managed-executor-service.grpc.core-thread | 16                                                          | Minimum number of Wildfly managed executors for gRPC service calls. (If it was not configured, it would be dynamic - [number of cpu cores] x 2)                                                                                                                    |
| OPENTELEMETRY_SERVICE_NAME                 | opentelemetry.service-name                | ROOT.war                                                    | Service name reported to the trace collector                                                                                                                                                                                                                       |
| OPENTELEMETRY_EXPORTER_TYPE                | opentelemetry.exporter-type               | jaeger                                                      | The name of the exporter to use: <ul style="list-style-type: none"> <li>• jaeger</li> <li>• otlp</li> </ul>                                                                                                                                                        |
| OPENTELEMETRY_EXPORTER_ENDPOINT            | opentelemetry.exporter.endpoint           | <a href="http://localhost:14250">http://localhost:14250</a> | The URL for the exporter endpoint.                                                                                                                                                                                                                                 |
| OPENTELEMETRY_SPAN_PROCESSOR_TYPE          | opentelemetry.span-processor.type         | batch                                                       | The type of span processor to use: <ul style="list-style-type: none"> <li>• batch: The default processor, which sends traces in batches as configured via the remaining attributes</li> <li>• simple: Traces are pushed to the exporter as they finish.</li> </ul> |
| OPENTELEMETRY_SPAN_PROCESSOR_BATCH_DELAY   | opentelemetry.span-processor.batch-delay  | 5000                                                        | The interval, in milliseconds, between two consecutive exports.                                                                                                                                                                                                    |

| ENV kulcs                                          | wildfly.properties kulcs                           | Default                                                                                                     | Leírás                                                                                                                                                                                                                                                      |
|----------------------------------------------------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OPENTELEMETRY_SPAN_PROCESSOR_MAX_QUEUE_SIZE        | opentelemetry.span-processor.max-queue-size        | 2048                                                                                                        | The maximum size of the queue before traces are dropped                                                                                                                                                                                                     |
| OPENTELEMETRY_SPAN_PROCESSOR_MAX_EXPORT_BATCH_SIZE | opentelemetry.span-processor.max-export-batch-size | 512                                                                                                         | The maximum number of traces that are published in each batch, which must be smaller or equal to <b>max-queue-size</b>                                                                                                                                      |
| OPENTELEMETRY_SPAN_PROCESSOR_EXPORT_TIMEOUT        | opentelemetry.span-processor.export-timeout        | 30000                                                                                                       | The maximum amount of time in milliseconds to allow for an export to complete before being cancelled                                                                                                                                                        |
| OPENTELEMETRY_SAMPLER_RATIO                        | opentelemetry.sampler.ratio                        | 0.01                                                                                                        | The value used to configure the ratio sampler, which must be within [0.0, 1.0]. For example, if 1 trace in 10,000 is to be exported, this value would be 0.0001.                                                                                            |
| OPENTELEMETRY_SAMPLER_TYPE                         | opentelemetry.sampler.type                         | ratio                                                                                                       | <p>The type of sampler to use:</p> <ul style="list-style-type: none"> <li>• on: Always on (all traces are recorded)</li> <li>• off: Always off (no traces are recorded)</li> <li>• ratio: Return a ratio of the traces (e.g., 1 trace in 10000).</li> </ul> |
| MICROMETER_ENDPOINT                                | micrometer.endpoint                                | <a href="http://opentelemetry-collector:4318/v1/metrics">http://opentelemetry-collector:4318/v1/metrics</a> | The URL to which metrics will be published.                                                                                                                                                                                                                 |
| MICROMETER_STEP                                    | micrometer.step                                    | 60                                                                                                          | The step size, or reporting frequency, to use (in seconds).                                                                                                                                                                                                 |

| ENV kulcs                            | wildfly.properties kulcs    | Default         | Leírás                                                                                                                                                                                                                                                               |
|--------------------------------------|-----------------------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HEALTH_EMPTY_LIVENESS_CHECKS_STATUS  | -                           | UP              | If the attribute is UP (by default), the server can be live when there are no liveness checks in the deployments. Setting the empty-liveness-checks-status attribute to DOWN will make this procedure fail when there are no liveness checks in the deployments.     |
| HEALTH_EMPTY_READINESS_CHECKS_STATUS | -                           | UP              | If the attribute is UP (by default), the server can be ready when there are no readiness checks in the deployments. Setting the empty-readiness-checks-status attribute to DOWN will make this procedure fail when there are no readiness checks in the deployments. |
| HEALTH_EMPTY_STARTUP_CHECKS_STATUS   | -                           | UP              | If the attribute is UP (by default), the server can be ready when there are no startup checks in the deployments. Setting the empty-startup-checks-status attribute to DOWN will make this procedure fail when there are no readiness checks in the deployments.     |
| UNDERTOW_HTTP_MAX_POST_SIZE          | undertow.http.max-post-size | 10485760 (10MB) | Size limit of HTTP POST requests. Larger sizes are automatically dropped by Wildfly and returned with an error message.                                                                                                                                              |

#### NOTE

The value of the key defined in `wildfly.properties` has been overridden by the value

of the key defined in ENV

## 2.3. wildfly/oracle-jdbc

Expands the common image created by [wildfly/microprofile](#) with oracle feature.

In this case, these images are created from it:

- **icellmobilsoft/wildfly-mp-jpa-oracle-jdbc:\${VERSION}** - **icellmobilsoft/wildfly-mp-jpa:\${VERSION}** + oracle driver
- **icellmobilsoft/wildfly-mp-jpa-full-oracle-jdbc:\${VERSION}** - **icellmobilsoft/wildfly-mp-full:\${VERSION}** + oracle driver

Configuration options are inherited from the parent [wildfly/microprofile](#) image and supplemented with the following:

Table 3. Configuration options:

| ENV kulcs                             | wildfly.properties kulcs             | Default                             | Leírás                                                                                     |
|---------------------------------------|--------------------------------------|-------------------------------------|--------------------------------------------------------------------------------------------|
| SPEC_DESCRIPTOR_PROPERTY_REPLACEMENT  | spec-descriptor.property-replacement | true                                | Use variables in the property/descriptor files (ejb-jar.xml, persistence.xml)              |
| ORACLE_DS_SERVICENAME                 | oracle.ds.servicename                | icellmobilsoft                      | Service name for datasource, postfixed with DS automatically → icellmobilsoftDS by default |
| ORACLE_DS_CONNECTION_URL              | oracle.ds.connection.url             | -                                   | Oracle DB connection url                                                                   |
| ORACLE_DS_USERNAME                    | oracle.ds.username                   | -                                   | Oracle DB connection user                                                                  |
| ORACLE_DS_PASSWORD                    | oracle.ds.password                   | -                                   | Oracle DB connection password                                                              |
| ORACLE_DS_NEW_CONNECTION_SQL          | oracle.ds.new-connection-sql         | BEGIN NULL; END;                    | Oracle DB new connection sql                                                               |
| ORACLE_DS_MIN_POOL_SIZE               | oracle.ds.min-pool-size              | 1                                   | Oracle DB connection min pool size                                                         |
| ORACLE_DS_MAX_POOL_SIZE               | oracle.ds.max-pool-size              | 60                                  | Oracle DB connection max pool size                                                         |
| DEFAULT_PERSISTENCE_HIBERNATE_DIALECT | -                                    | org.hibernate.dialect.OracleDialect | Implementation constant value for usage in persistence.xml file                            |

**NOTE**

The value of the key defined in **wildfly.properties** has been overridden by the value of the key defined in **ENV**

Example for DEFAULT\_PERSISTENCE\_HIBERNATE\_DIALECT usage: [persistence.xml](#)

## 2.4. wildfly/postgresql-jdbc

Expands the common image created by [wildfly/microprofile](#) with postgresql feature.

In this case, these images are created from it:

- **icellmobilsoft/wildfly-mp-jpa-postgresql-jdbc:\${VERSION}** - **icellmobilsoft/wildfly-mp-jpa:\${VERSION}** + postgresql driver
- **icellmobilsoft/wildfly-mp-full-postgresql-jdbc:\${VERSION}** - **icellmobilsoft/wildfly-mp-full:\${VERSION}** + postgresql driver

Configuration options are inherited from the parent [wildfly/microprofile](#) image and supplemented with the following:

Table 4. Configuration options:

| ENV kulcs                            | wildfly.properties kulcs             | Default        | Leírás                                                                                     |
|--------------------------------------|--------------------------------------|----------------|--------------------------------------------------------------------------------------------|
| SPEC_DESCRIPTOR_PROPERTY_REPLACEMENT | spec-descriptor.property-replacement | true           | Use variables in the property/descriptor files (ejb-jar.xml, persistence.xml)              |
| POSTGRESQL_DS_SERVICE_NAME           | postgresql.ds.service name           | icellmobilsoft | Service name for datasource, postfixed with DS automatically → icellmobilsoftDS by default |
| POSTGRESQL_DS_CONNECTION_URL         | postgresql.ds.connection.url         | -              | PostgreSQL DB connection url                                                               |
| POSTGRESQL_DS_USERNAME               | postgresql.ds.username               | -              | PostgreSQL DB connection user                                                              |
| POSTGRESQL_DS_PASSWORD               | postgresql.ds.password               | -              | PostgreSQL DB connection password                                                          |
| POSTGRESQL_DS_NEW_CONNECTION_SQL     | postgresql.ds.new-connection-sql     | BEGIN;END;     | PostgreSQL DB new connection sql                                                           |
| POSTGRESQL_DS_MIN_POOL_SIZE          | postgresql.ds.min-pool-size          | 1              | PostgreSQL DB connection min pool size                                                     |

| ENV kulcs                             | wildfly.properties kulcs    | Default                                 | Leírás                                                          |
|---------------------------------------|-----------------------------|-----------------------------------------|-----------------------------------------------------------------|
| POSTGRESQL_DS_MAX_POOL_SIZE           | postgresql.ds.max-pool-size | 60                                      | PostgreSQL DB connection max pool size                          |
| DEFAULT_PERSISTENCE_HIBERNATE_DIALECT | -                           | org.hibernate.dialect.PostgreSQLDialect | Implementation constant value for usage in persistence.xml file |

#### NOTE

The value of the key defined in **wildfly.properties** has been overridden by the value of the key defined in **ENV**

Example for DEFAULT\_PERSISTENCE\_HIBERNATE\_DIALECT usage: [persistence.xml](#)

## 2.5. wildfly/ibm-mq

Expands the common image created by [wildfly/oracle-jdbc](#) with ibm-mq feature.

In this case, these images are created from it:

- **icellmobilsoft/wildfly-mp-full-oracle-jdbc-ibm-mq:\${VERSION}** - **icellmobilsoft/wildfly-mp-full-oracle-jdbc:\${VERSION}** + ibm-mq driver

## 2.6. wildfly/mariadb-jdbc

Expands the common image created by [wildfly/microprofile](#) with mariadb feature.

In this case, these images are created from it:

- **icellmobilsoft/wildfly-mp-jpa-mariadb-jdbc:\${VERSION}** - **icellmobilsoft/wildfly-mp-jpa:\${VERSION}** + mariadb driver
- **icellmobilsoft/wildfly-mp-full-mariadb-jdbc:\${VERSION}** - **icellmobilsoft/wildfly-mp-full:\${VERSION}** + mariadb driver

Configuration options are inherited from the parent [wildfly/microprofile](#) image and supplemented with the following:

Table 5. Configuration options:

| ENV kulcs                            | wildfly.properties kulcs             | Default | Leírás                                                                        |
|--------------------------------------|--------------------------------------|---------|-------------------------------------------------------------------------------|
| SPEC_DESCRIPTOR_PROPERTY_REPLACEMENT | spec-descriptor.property-replacement | true    | Use variables in the property/descriptor files (ejb-jar.xml, persistence.xml) |

| ENV kulcs                             | wildfly.properties kulcs      | Default                              | Leírás                                                                                     |
|---------------------------------------|-------------------------------|--------------------------------------|--------------------------------------------------------------------------------------------|
| MARIADB_DS_SERVICE_NAME               | mariadb.ds.servicename        | icellmobilsoft                       | Service name for datasource, postfixed with DS automatically → icellmobilsoftDS by default |
| MARIADB_DS_CONNECTION_URL             | mariadb.ds.connection.url     | -                                    | MariaDb DB connection url                                                                  |
| MARIADB_DS_USERNAME                   | mariadb.ds.username           | -                                    | MariaDb DB connection user                                                                 |
| MARIADB_DS_PASSWORD                   | mariadb.ds.password           | -                                    | MariaDb DB connection password                                                             |
| MARIADB_DS_NEW_CONNECTION_SQL         | mariadb.ds.new-connection-sql | SELECT SQL_NO_CACHE 1                | MariaDb DB new connection sql                                                              |
| MARIADB_DS_MIN_POOL_SIZE              | mariadb.ds.min-pool-size      | 1                                    | MariaDb DB connection min pool size                                                        |
| MARIADB_DS_MAX_POOL_SIZE              | mariadb.ds.max-pool-size      | 60                                   | MariaDb DB connection max pool size                                                        |
| DEFAULT_PERSISTENCE_HIBERNATE_DIALECT | -                             | org.hibernate.dialect.MariaDBDialect | Implementation constant value for usage in persistence.xml file                            |

#### NOTE

The value of the key defined in **wildfly.properties** has been overridden by the value of the key defined in **ENV**

Example for DEFAULT\_PERSISTENCE\_HIBERNATE\_DIALECT usage: [persistence.xml](#)

## 3. JMX usage

The modules for the wildfly jmx subsystem will be included in the images created by galleon, but by default the subsystem is disabled. On developer images it can be activated with a cli script, so there is no need to run a separate shell script for jmx.

*configure-jmx.cli*

```
/subsystem=jmx:add
/subsystem=jmx/remoting-connector=jmx:add(use-management-endpoint=true)

# management auth is turned off - not important for local execution anyway
/core-service=management/management-interface=http-interface:undefine-attribute(name=http-authentication-factory)
/core-service=management/management-interface=http-interface:map-remove(name=http-
```

```
upgrade, key=sasl-authentication-factory)
```

To connect, you may need wildfly-cli.jar on the client classpath. This can be done for example as follows:

- download client jar

```
mvn org.apache.maven.plugins:maven-dependency-plugin:2.8:get
-Dartifact=org.wildfly.core:wildfly-cli:19.0.1.Final:jar:client
```

- add the downloaded jar to the classpath when starting the jmx client

*visualvm sample*

```
visualvm --cp:a ${HOME}/.m2/repository/org/wildfly/core/wildfly-
cli/19.0.1.Final/wildfly-cli-19.0.1.Final-client.jar
```

*jconsole sample*

```
jconsole -J
-Djava.class.path="${JAVA_HOME}/lib/jconsole.jar:${JAVA_HOME}/lib/tools.jar:${HOME}
/.m2/repository/org/wildfly/core/wildfly-cli/19.0.1.Final/wildfly-cli-19.0.1.Final-
client.jar"
```

### IMPORTANT

for the above commands, you should make sure that the environment variables used exist, or replace them with absolute paths, ~ cannot be used for user home!

Wildfly allows jmx connection on the management port, url must be in the format `service:jmx:remote+http://localhost:${management.port}`.

## 4. Examples

*persistence.xml*

```
<?xml version="1.0" encoding="UTF-8"?>
<persistence xmlns="https://jakarta.ee/xml/ns/persistence"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="https://jakarta.ee/xml/ns/persistence
    https://jakarta.ee/xml/ns/persistence/persistence_3_0.xsd"
  version="3.0">

  <persistence-unit name="defaultPU" transaction-type="JTA">
    ...
    <properties>
      ...
      <property name="hibernate.dialect"
```

```
value="${env.DEFAULT_PERSISTENCE_HIBERNATE_DIALECT}" />
    </properties>
</persistence-unit>
</persistence>
```

*docker-compose.yml*

```
...
  environment:
    DEBUG: true
    #different gc
    JAVA_GC: '-XX:+UseSerialGC'
    #JVM default gc with empty string:
    #JAVA_GC: ''
  ports:
...

```

## 5. Release Notes

### 5.1. v2.0.0

The project has moved out of the internal repository, not counting updates, it is 1:1 compatible with version 1.1.0.

*Changes compared to the internal version*

- Renames:
  - wildfly-ms/default → icellmobilsoft/wildfly-default-ms
  - wildfly-jpa/default → icellmobilsoft/wildfly-default-jpa
  - wildfly-full/default → icellmobilsoft/wildfly-default-full
  - wildfly-ms/mp → icellmobilsoft/wildfly-mp-ms
  - wildfly-ms/jpa/mp → icellmobilsoft/wildfly-mp-jpa
  - wildfly-full/mp → icellmobilsoft/wildfly-mp-full
  - wildfly-ms/logstash → icellmobilsoft/wildfly-mp-ms
  - wildfly-ms/jpa-logstash → icellmobilsoft/wildfly-mp-jpa
  - wildfly-full/logstash → icellmobilsoft/wildfly-mp-full
  - wildfly-ms/oracle-jdbc → icellmobilsoft/wildfly-mp-jpa-oracle-jdbc
  - wildfly-full/oracle-jdbc → icellmobilsoft/wildfly-mp-full-oracle-jdbc
  - wildfly-ms/postgresql-jdbc → icellmobilsoft/wildfly-mp-jpa-postgresql-jdbc
  - wildfly-full/postgresql-jdbc → icellmobilsoft/wildfly-mp-full-postgresql-jdbc
  - wildfly-full/oracle-jdbc-ibm-mq → icellmobilsoft/wildfly-mp-full-oracle-jdbc-ibm-mq
  - (replaced) wildfly-ms/gcp → icellmobilsoft/wildfly-mp-ms

- (replaced) wildfly-ms/jpa-gcp → icellmobilsoft/wildfly-mp-jpa
- (replaced) wildfly-full/gcp → icellmobilsoft/wildfly-mp-full
- (replaced) wildfly-ms/postgresql-jdbc-gcp → icellmobilsoft/wildfly-mp-jpa-postgresql-jdbc
- (replaced) wildfly-full/postgresql-jdbc-gcp → icellmobilsoft/wildfly-mp-full-postgresql-jdbc
- Environment changes:
  - (new with old value) `ORACLE_DS_SERVICENAME` default `icellmobilsoft+DS`
  - (new with old value) `POSTGRESQL_DS_CONNECTION_URL` default `icellmobilsoft+DS`
  - (new) `CONSOLE_LOGGING_JSON_ENABLED` default `false`
  - (new) `CONSOLE_LOGGING_JSON_FORMATTER_ZONEID` default `UTC`
  - (change value) `FILE_LOGGING_ENABLED` default `false` (changed from `true`)
  - (new with old value) `LOGSTASH_BLOCKING_MODE` default `DISCARD`
  - (new with old value) `LOGSTASH_QUEUE_LENGTH` default `50000`
  - (change value) `LOGSTASH_ENABLED` default `false` (changed from `true`)
- Debugging: The `--debug` option doesn't need anymore in the `CMD exec` which you can ignore in the local Dockerfile mostly. The debug mode can be controlled by the `DEBUG` environment variable which is false by default. The default port is `8787` but you can override it with the `DEBUG_PORT` environment variable.

#### WARNING

Don't release docker images with `DEBUG` enabled into production environments!

#### What's new

- Baseimage version update to 1.4.0
- `DEFAULT_PERSISTENCE_HIBERNATE_DIALECT` variable added to the jdbc images to use in the local Dockerfiles:
  - wildfly-mp-\*-oracle-jdbc image = `org.hibernate.dialect.OracleDialect`
  - wildfly-mp-\*-postgresql-jdbc image = `org.hibernate.dialect.PostgreSQLDialect`

#### NOTE

*project-local.Dockerfile snippet*

You can override default dialect with `PERSISTENCE_HIBERNATE_DIALECT` environment variable but you dont need to set it in the `docker-compose.yml` file mostly.

ARG

`PERSISTENCE_HIBERNATE_DIALECT=$DEFAULT_PERSISTENCE_HIBERNATE_DIALECT ENV HIBERNATE_DIALECT=$PERSISTENCE_HIBERNATE_DIALECT`

## 5.2. v2.0.1

#### What's new

- ironjacamar auto commit disabled when stopping container

- Set the default garbage collector to g1 gc. With JAVA\_GC environment variable the default gc can be overridden. Just set JAVA\_GC env variable in docker-compose file. Correct values: '-XX:+UseConcMarkSweepGC' ; '-XX:+UseSerialGC' ; '-XX:+UseG1GC' etc

## 5.3. v3.0.0

*What's new*

- Bump Wildfly 27.0.1 → 30.0.1

*Changes compared to previous version*

- Replace [Microprofile Opentracing](#) with [Microprofile Telemetry 1.1 Specification](#)
- Replace [Microprofile Metrics](#) with [Micrometer](#)
- Environment changes:
  - (removed) `OPENTRACING_JAEGER_AGENT_HOST`
  - (removed) `OPENTRACING_JAEGER_AGENT_PORT`
  - (removed) `OPENTRACING_JAEGER_AGENT_SAMPLER_PORT`
  - (removed) `OPENTRACING_JAEGER_PROPAGATION`
  - (new) `MICROMETER_ENDPOINT` default: <http://opentelemetry-collector:4318/v1/metrics>
  - (new) `MICROMETER_STEP` default: `60`
  - (new) `OPENTELEMETRY_SERVICE_NAME` default: `ROOT.war`
  - (new) `OPENTELEMETRY_EXPORTER_TYPE` default: `jaeger`
  - (new) `OPENTELEMETRY_EXPORTER_ENDPOINT` default: <http://localhost:14250>
  - (new) `OPENTELEMETRY_SPAN_PROCESSOR_TYPE` default: `batch`
  - (new) `OPENTELEMETRY_SPAN_PROCESSOR_BATCH_DELAY` default: `5000`
  - (new) `OPENTELEMETRY_SPAN_PROCESSOR_MAX_QUEUE_SIZE` default: `2048`
  - (new) `OPENTELEMETRY_SPAN_PROCESSOR_MAX_EXPORT_BATCH_SIZE` default: `512`
  - (new) `OPENTELEMETRY_SPAN_PROCESSOR_EXPORT_TIMEOUT` default: `30000`
  - (new) `OPENTELEMETRY_SAMPLER_RATIO` default: `0.01`
  - (new) `OPENTELEMETRY_SAMPLER_TYPE` default: `ratio`
- Add new wildfly jpa base images with mariaDb support, see [wildfly/mariadb-jdbc](#)
  - icellmobilsoft/wildfly-mp-jpa-mariadb-jdbc
  - icellmobilsoft/wildfly-mp-full-mariadb-jdbc

**NOTE** | See configuration at [here](#)

## 5.4. v3.0.1

*Changes compared to previous version*

- Upgrade Oracle JDBC version from 23.3 to 23.4
- Separate managed-executor-service (java:jboss/ee/concurrency/executor/grpc) for gRPC threads

## 5.5. v3.1.0

*Changes compared to previous version*

- GH action: bake-action upgrade docker/bake-action@v4 → docker/bake-action@v5
- JSON logging enhancements:
  - `biz.paluch.logging.gelf.wildfly.WildFlyJsonFormatter` configured supporting our standard logging format
    - Environment changes:
      - (removed) `CONSOLE_LOGGING_JSON_FORMATTER_ZONEID` default `UTC`
        - with `WildFlyJsonFormatter` there is no option to set the timezone, thus adding zone offset to timestamp: `yyyy-MM-dd HH:mm:ss,SSSZ`
      - (new) `JSON_MODULE_ID` default `unknown`
      - (new) `JSON_MODULE_VERSION` default `unknown`
      - (new) `JSON_K8S_NAMESPACE` default `unknown`
      - (new) `JSON_MDC_FIELDS`

## 5.6. v4.0.0

*Changes compared to previous version*

- Opentelemetry default otlp exporter type, jaeger is not supported
- galleon 6+ missing user home workaround
- base image upgrade 1.5.0
- WildFly 34 upgrade
- JSON logging:
  - Normal WF JSON logging restored:
    - GELF formatting is removed
    - can be activated with `CONSOLE_LOGGING_JSON_ENABLED`
    - Environment variables can be used to set additional fields in the JSON log:
      - `JSON_MODULE_ID` default `unknown`
      - `JSON_MODULE_VERSION` default `unknown`
      - `JSON_K8S_NAMESPACE` default `unknown`
      - Info: adding additional mdc fields via env is not supported by the formatter meta-data configuration
  - GELF JSON logging separated from normal WF JSON logging:

- can be activated with new env: `CONSOLE_LOGGING_GELF_JSON_ENABLED`
- Environment changes:
  - (new) `GELF_JSON_MODULE_ID` default `unknown`
  - (new) `GELF_JSON_MODULE_VERSION` default `unknown`
  - (new) `GELF_JSON_K8S_NAMESPACE` default `unknown`
  - (new) `GELF_JSON_MDC_FIELDS`

## 5.7. v5.0.0

*Changes compared to previous version*

- base image upgrade 1.8.0
- WildFly 38 upgrade

## 5.8. v5.1.0

*Changes compared to previous version*

- Base image version upgraded from `1.8.0` to `1.9.0`. Notable changes:
  - base image is upgraded from `debian:12.12-slim` to `debian:12.15-slim`
  - JRE and JDK 21 `21.0.9_10` → `21.0.11_10`
  - galleon upgraded from `6.1.1.Final` → `7.0.8.Final`
- Removed `git-history` galleon layer from generated WildFly

*Maintenance*

- Extracted `wildfly-mp-full-oracle-jdbc-ibm-mq-release` image from `docker-compose-3.yml` to a new file: `docker-compose-4.yml`